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## Stage 1: Patient information input

### Patient Information

**Basic information and chief complaint:** The patient is an 88-year-old woman who was admitted to the hospital due to abdominal pain accompanied by vomiting for one week.
**History of present illness:** One week prior to admission, the patient developed abdominal pain without any obvious precipitating factors, accompanied by abdominal distension, nausea, and vomiting. The vomitus consisted of yellow, retained gastric contents. She also reported absence of bowel movements during this period. There were no associated symptoms such as chest tightness, chest pain, diarrhea, dizziness, headache, palpitations, or shortness of breath.
She initially sought medical care at another hospital, where laboratory tests revealed anemia and impaired renal function. As her abdominal pain persisted without significant relief, she subsequently presented to our hospital for further evaluation and management and was admitted with a preliminary diagnosis of a "colonic mass."
Since symptom onset, the patient has remained conscious and oriented, with a fair mental status. She has had poor appetite and suboptimal sleep quality. Urination has been unremarkable, and there has been no significant change in body weight in the recent period.
**Complete blood count and coagulation profile:** Complete blood count showed: red blood cell count 2.32 × $10^{12}$/L, hemoglobin 72 g/L, white blood cell count 3.61 × $10^{9}$/L, neutrophils 2.1 × $10^{9}$/L, and platelet count 105 × $10^{9}$/L. Coagulation studies revealed a prothrombin time (PT) of 13.3 seconds and an activated partial thromboplastin time (APTT) of 30.4 seconds.
**Blood biochemistry and electrolytes:** Biochemical tests demonstrated a serum creatinine level of 166.0 μmol/L and a blood glucose level of 17.93 mmol/L. Prealbumin was 147 mg/L, albumin 38 g/L, and total protein 60 g/L. Electrolyte analysis showed hyponatremia (sodium 127.0 mmol/L) and hypomagnesemia (magnesium 0.49 mmol/L). Other electrolytes, including potassium, chloride, calcium, as well as liver function–related parameters, were within normal limits. CA125 (42.4 U/mL)
**Tumor marker assessment:** Tumor marker evaluation revealed a markedly elevated CA72-4 level of 74.53 U/mL. CA19-9 (40.1 U/mL) and were mildly elevated. Carcinoembryonic antigen (CEA), alpha-fetoprotein (AFP), and CA242 were within normal ranges.

## Stage 2: Endoscopy

*Action: recommend examination="endoscopy"*

### Endoscopy reports

**Endo-agent report:** Suspected cancer. A mass is observed in the colon, dark red in color, with mild surface erosion and irregular margins, causing luminal narrowing and obstruction. The surface of the lesion is uneven, and the surrounding mucosa is congested, with mild oozing of blood.

***Human-written report (reference):*** *The endoscope was advanced to the hepatic flexure. Diffuse colonic melanosis was noted, with abundant fecal fluid, stool, and fruit seeds partially obscuring the view. Hepatic flexure: A purple extrinsic compression mark was seen. A proliferative lesion obstructed the lumen, preventing scope passage; three biopsy specimens were obtained. Transverse colon: Normal architecture with a clear vascular pattern. Splenic flexure, descending colon, sigmoid colon: No obvious abnormalities. Rectum: Rectal ampulla present; mucosa smooth. Anal canal: No obvious abnormalities.*
*Impression: Proliferative lesion at the hepatic flexure–ascending colon with luminal narrowing (pathology pending); colonic melanosis.*

### <Reason> Summary of the patient's condition and analysis of relevant guideline-based recommendations

**Patient summary**

- The patient is an 88-year-old woman of advanced age. She presented with abdominal pain accompanied by vomiting for one week. The vomitus consisted of retained food, and symptoms were associated with abdominal distension and absence of bowel movements, suggesting features of intestinal obstruction. On admission, she was conscious and oriented, with fair condition, poor appetite, and suboptimal sleep.
- **Laboratory findings:** Significant anemia (Hb 72 g/L), with reduced white blood cell and platelet counts. Renal insufficiency was noted (creatinine 166 μmol/L, urea 5.5 mmol/L). Hyperglycemia consistent with diabetes mellitus was present (blood glucose 17.93 mmol/L). Hyponatremia was observed (Na 127.0 mmol/L). Low prealbumin levels indicated poor nutritional status. Among tumor markers, CA72-4 was markedly elevated (74.53 U/mL), while CA19-9 and CA125 were mildly elevated; the remaining markers were within normal limits.
- **Endoscopic findings:** The endoscopic appearance was consistent with typical features of early or advanced colorectal cancer, including foul odor, ulceration with necrosis, firm consistency, and ill-defined margins, supporting the diagnosis of a colonic mass (suspected malignancy). Mild bleeding (oozing) was also observed.
- **Completed evaluations:** Basic laboratory tests (complete blood count, biochemistry, coagulation profile, and tumor markers) and endoscopic examination have been completed. Based on the findings, malignancy is highly suspected, pending pathological confirmation.

**Guideline retrieval direction analysis**

- **Cancer diagnosis and staging guidelines:** Endoscopic findings strongly suggest colorectal malignancy. Definitive diagnosis requires histopathological confirmation. Subsequent management should be based on colorectal cancer staging according to the AJCC 8th edition TNM classification.
- **Imaging guidelines:** To achieve accurate staging, contrast-enhanced CT of the abdomen and pelvis should be performed, with chest CT if necessary, to evaluate local tumor extent, lymph node status, and distant metastasis. This approach is consistent with standard MDT requirements for colorectal cancer staging.
- **Guidelines for advanced age and comorbidity management:** Given the patient's advanced age (88 years) and multiple comorbidities—including anemia, renal insufficiency, diabetes mellitus, and electrolyte disturbances—a comprehensive geriatric oncology assessment is required prior to treatment planning. Concurrent standardized management of comorbid conditions is essential to ensure safety and tolerability.
- **Nutrition and supportive care guidelines:** The presence of anemia, low prealbumin levels, and poor appetite indicates nutritional risk. Nutritional assessment and appropriate interventions should be implemented in accordance with established guidelines for nutritional support in oncology patients, in parallel with tumor evaluation.

**Summary**

The patient is an elderly woman with multiple comorbidities and suspected colorectal cancer, with endoscopic findings highly suggestive of malignancy. The immediate priorities are histopathological confirmation via biopsy and contrast-enhanced CT imaging to determine TNM staging. A comprehensive evaluation incorporating geriatric oncology principles and comorbidity management is essential. Definitive treatment decisions should be individualized based on imaging results and final pathological diagnosis, in accordance with best practices for elderly patients.

*Action: get guideline keywords="Elderly patient with suspected complicated intestinal obstruction and possible malignancy, pathology pending."*

### Retrieved clinical guidelines

[1] Histologic confirmation is required before initiating definitive treatment, unless emergent surgery is required for life-threatening complications.
[2] Imaging findings should be interpreted in conjunction with endoscopic and pathologic information and reviewed in a multidisciplinary setting when possible.
[3] Contrast-enhanced CT of the chest, abdomen, and pelvis is recommended to evaluate the extent of locoregional disease, assess regional lymph node involvement, and detect distant metastases, particularly to the liver, lungs, and peritoneum.

### Recommending examination: pathology

The patient is an 88-year-old woman who presented with abdominal pain accompanied by vomiting for one week and a fair general condition. Her medical history is notable for diabetes mellitus, moderate anemia, and impaired renal function. She has experienced difficulty with oral intake recently and a trend toward weight loss, indicating poor overall physical status and a risk of malnutrition. Endoscopic examination revealed mucosal erosion and features suggestive of malignant transformation, raising suspicion for colorectal malignancy with concomitant signs of intestinal obstruction. A treatment plan should be formulated after confirming the diagnosis and delineating the extent of disease, with careful consideration of the patient's advanced age and comorbidities and their impact on treatment tolerability.

Based on the above assessment, to clarify disease staging and the site of obstruction, the following evaluations should be further completed:

- **Pathological examination:** to confirm the diagnosis of cancer [1,2].
- **Contrast-enhanced CT / virtual endoscopy:** to further define the extent of disease (contrast use requires careful assessment of renal function, particularly in patients with renal impairment) [3].
- **Blood and urine biochemical monitoring:** to ensure that electrolyte abnormalities have not worsened and to closely monitor serum creatinine and renal function.
- **Repeat tumor marker testing:** such as CEA, to assess tumor burden and disease progression.

*Action: recommend examination="pathology"*

**Please refer to the next page for subsequent stages.**

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**Continued from the previous page**

## Stage 3: Pathology

*Action: recommend examination="pathology"*

**Pathology reports**

**Patho-agent diagnosis:** Cancer

***Human-written report (reference):*** *(Hepatic flexure biopsy) Adenocarcinoma, with inflammatory changes and calcified old schistosome eggs.*

**<Reason> Summary of the patient's condition and analysis of relevant guideline-based recommendations**

**Patient summary:** (1) **Patient condition, laboratory findings, endoscopic findings:** refer to the reasoning in the endoscopy stage; (2) **Pathological findings:** diagnosed as cancer.

**Guideline retrieval direction analysis:** The patient has been definitively diagnosed with colon cancer, accompanied by symptoms of intestinal obstruction as well as systemic conditions such as anemia and impaired renal function. The next steps in diagnostic and therapeutic decision-making should primarily be guided by the following clinical guidelines:

- **Guidelines for the diagnosis and treatment of colorectal cancer:** Authoritative guidance covering the diagnosis, staging, and treatment of colorectal cancer.
- **Cancer staging guidelines:** refer to the reasoning in the endoscopy stage.
- **Expert consensus on anesthesia and perioperative management of elderly oncology patients:** Given the patient's advanced age (88 years) and multiple comorbid conditions—including anemia, renal insufficiency, hyperglycemia, and electrolyte disturbances—she is at high perioperative risk.
- **Guidelines for nutritional support and management in patients with colorectal cancer:** refer to the reasoning in the endoscopy stage.

**Summary:** The patient is an elderly individual with colon cancer who has developed intestinal obstruction and complications, with pathological confirmation of malignancy. The key of subsequent management is to determine the treatment strategy based on the stage of cancer, while fully assessing the surgical and therapeutic risks associated with advanced age and prioritizing evaluation and support of nutritional status.

*Action: get guideline keywords="Elderly patient with colon cancer with intestinal obstruction, anemia, and renal insufficiency."*

**Retrieved clinical guidelines**

[1] All patients with confirmed colorectal cancer should undergo clinical staging prior to treatment planning whenever feasible…
[2] Imaging findings should be interpreted in conjunction with endoscopic and pathologic information and reviewed in a multidisciplinary setting when possible.
[3] Optimization of nutritional status is recommended prior to major surgical interventions whenever possible, as…
[4] The geriatric assessment should evaluate functional status, comorbidities, cognition, nutritional status, renal function, metabolic abnormalities…

**Recommending examination: CT**

The patient is an 88-year-old woman with a confirmed diagnosis of colon cancer … (repeated details of the clinical summary are omitted here).

**Guideline retrieval direction analysis**

- **Chest, abdominal, and pelvic CT examination [1,2]:** To determine accurate tumor staging, including the presence of lymph node involvement or distant metastases. This evaluation is critical for guiding subsequent treatment decisions. Decisions regarding surgery versus chemotherapy should be based on CT findings, with assessment of the patient's physiological tolerance given her advanced age.
- **Preoperative preparation [3,4]:** Correction of anemia and electrolyte disturbances, optimization of glycemic control, and provision of enteral or parenteral nutritional support to improve overall nutritional status. Careful medication selection is required to avoid nephrotoxic agents, with close monitoring of renal function.
- **Points of consideration:** The treatment plan should be formulated in conjunction with the patient and her family to ensure alignment with the patient's overall best interests and quality of life.

## Stage 4: CT

*Action: recommend examination=CT"*

**CT reports**

**Radio-agent report:** There is marked, irregular thickening of the bowel wall in the ascending colon and near the hepatic flexure, with luminal narrowing, obstruction, and infiltration of the surrounding fat. Obvious heterogeneous enhancement is observed in this region. In addition, small lymph nodes are noted in the adjacent mesentery. No abnormally enlarged lymph nodes are seen in the retroperitoneal region.

***Human-written report (reference):*** *No definite calculi are identified, and no abnormal calcifications are observed. The urinary bladder is normal in contour, with a smooth wall and no abnormal intraluminal densities. No significantly enlarged lymph nodes are seen in the retroperitoneum. A high-density nodular focus is noted within the uterus, likely representing calcification; no obvious abnormalities are seen in the bilateral adnexal regions. Pelvic free fluid is present. A space-occupying lesion is seen in the ascending colon with suspicious surrounding lymph nodes. Uterine calcification and pelvic effusion are noted. Correlation with clinical findings and follow-up with relevant examinations are recommended.*
***Human-assessed clinical TNM stage (reference):*** *cT3N+M0*
***Human-written postoperative pathology (reference):*** *(pT4aN1bM0) The right hemicolectomy specimen confirmed a poorly differentiated mucinous adenocarcinoma of the ascending colon (with focal signet ring cell carcinoma). The tumor invaded the serosal layer and showed marked lymphovascular and perineural invasion, with lymph node metastases (3/14) and cancer nodules close to the resection margin, indicating advanced-stage colon cancer.*

**<Reason> Summary of the patient's condition and analysis of relevant guideline-based recommendations**

**Patient summary:** (1) **Patient condition, laboratory findings, endoscopic and pathological findings:** refer to the reasoning in the pathology stage; (2) **CT findings**: Irregular thickening of the bowel wall in the ascending colon and near the hepatic flexure, with luminal narrowing, infiltration of the surrounding fat, and enlargement of adjacent mesenteric lymph nodes (suggestive of possible nodal metastasis). No abnormal enlargement of retroperitoneal lymph nodes is observed.
Overall, the patient is an elderly woman presenting with symptoms of intestinal obstruction. Imaging and pathological examinations have confirmed the diagnosis of colon cancer, and CT findings suggest possible regional lymph node metastasis. The preliminary clinical TNM stage is assessed as at least T3N1M0.

**Guideline retrieval direction analysis:** Guidelines for the treatment of colon cancer; Guidelines for geriatric oncology patients; Guidelines for the treatment of intestinal obstruction…(refer to the pathology stage)

**Summary:** All essential investigations and clinical staging analyses have been completed. Further management should strictly follow guidelines for the care of geriatric oncology patients, including comprehensive assessment of functional and physiological status, as well as guidelines for the management of intestinal obstruction to address urgent symptoms. Based on these considerations, appropriate subsequent treatment strategies—such as palliative care, decompression, and supportive therapy—should be recommended.

*Action: get guideline keywords="Elderly woman with colon cancer presenting with abdominal pain, vomiting, and intestinal obstruction, accompanied by anemia, hyponatremia, and impaired renal function; CT shows lymph node metastasis"*

**Retrieved clinical guidelines**

[1] Surgical resection is the primary treatment for localized colon cancer and may be performed with curative or palliative intent depending on clinical circumstances…Colon cancer surgery can be classified by tumor location and procedure type, including left hemicolectomy, right hemicolectomy, sigmoid colectomy, and rectal surgery. For colon cancer, commonly used chemotherapy regimens include XELOX, FOLFOX, and FOLFIRI.
[2] Patients with colon cancer who present with bowel obstruction…treatment options may include: emergent or urgent surgical resection diverting ostomy, or endoscopic stent placement… In elderly patients or those with poor performance status and significant comorbidities, less invasive approaches may be preferred...
[3] In patients with poor performance status or severe comorbidities, less intensive or palliative treatment strategies may be more appropriate, with an emphasis on symptom control and quality of life.
[4] Optimization of nutritional status is recommended prior to major surgical interventions whenever possible…

*Action: recommend treatment*

## Stage 5: Treatment

**Final treatment recommendation**

The patient is an 88-year-old woman who presented with abdominal pain accompanied by vomiting for one week. Clinically, a diagnosis of locally advanced ascending colon cancer involving the region near the hepatic flexure is suspected, with possible regional lymph node metastasis but no clear evidence of distant metastasis. The preliminary clinical TNM stage is assessed as T3N1M0. The patient is currently experiencing intestinal obstruction and has significant laboratory abnormalities, including severe anemia (hemoglobin 72 g/L), impaired renal function (creatinine 166.0 μmol/L), and electrolyte disturbances (hyponatremia and hyperglycemia). Although she is mostly conscious and oriented, her overall performance status is poor, and she may have limited tolerance for high-intensity surgery or chemotherapy.

**Treatment recommendation**

- **Surgical option:** If the patient has a strong desire for active treatment and her overall condition permits, right hemicolectomy [1] is recommended as the surgical option (with individualized consideration of perioperative or intraoperative chemotherapy where appropriate). Given her advanced age, the risks associated with anesthesia and surgery are high, and a comprehensive preoperative assessment is required (e.g., cardiopulmonary function and gastrointestinal obstruction symptoms). Either a minimally invasive or open surgical approach may be considered based on intraoperative and preoperative factors.
- **Non-surgical options:** Considering the patient's advanced age, if she is deemed unable to tolerate surgery after evaluation (e.g., due to poor cardiopulmonary function), curative surgery should be avoided, and palliative treatment is preferred [3]. This may include palliative chemoradiotherapy for tumor control or creation of a distal gastrointestinal stoma to relieve intestinal obstruction [2].
- **Chemotherapy and radiotherapy**: FOLFOX [1] is the standard chemotherapy regimen for colorectal cancer. Considering the patient's low tolerance (age and renal function), a less intensive option such as XELODOC (tegafur monotherapy) may be considered [1], with dose reduction to approximately 60%–75%. Targeted therapy (e.g., bevacizumab) may be considered if unresectable and molecular testing permits.
- **Palliative care and nutritional support**: (1) Nutritional support [4]: Abdominal pain and distension are primarily related to intestinal obstruction and impaired bowel transit. Gastrointestinal decompression and nutritional support are required. Stoma decompression may be considered to further relieve abdominal symptoms. (2) Psychological support: Given the patient's age and the psychological burden of a cancer diagnosis, both the patient and her family may benefit from psychological counseling to establish realistic treatment goals focused on maximizing quality of life and functional well-being. (3) Ongoing follow-up during chemotherapy: Close monitoring of complete blood counts and biochemical parameters is essential, with timely adjustment of treatment strategies as needed.

**Humanistic care:** Enhanced nutritional support and psychological counseling should be emphasized in elderly patients, with quality of life as the primary goal. Prognosis and treatment expectations should be thoroughly discussed with the family to clarify goals of care. Based on clinical guidelines and the patient's actual condition, decisions regarding chemotherapy or surgery should be precisely tailored after complete staging. This is the objective of the multidisciplinary medical team. The cooperation of the patient's family is greatly appreciated, and every effort will be made to provide comprehensive support for the patient.